Milestone 7: Star Value Model

What one extra care-gap closure is worth through Star Ratings and QBP

Prepared by: KSU mHealth Research Lab | Date: 2026-10-10

RESEARCH FRAMEWORK. Lift is research-generated or taken from published trials; no InComm lift, financial or issuance data is used. Dollar values are research ranges tagged by provenance.

1. Method

- Value of a closure = QBP dollars per enrollee x boundary density (rho) x pass-through / eligible share / gaps sharing the measure.
- rho is measured, not assumed: for every rated Medicare Advantage contract in CMS's public 2026 Star Ratings data, how likely is a small rate increase to cross a cut point and lift the overall rating over the 4-star QBP boundary.
- QBP per enrollee: KFF 2026, $318 / $381 / $466. Eligible shares: public prevalence anchors (diabetes 0.20 / 0.26 / 0.32; hypertension 0.50 / 0.60 / 0.65).

2. Measured boundary density (CMS 2026, public)

Measure ID	Weight	N Contracts	Mean Rate	Rho Low	Rho Base	Rho High
C11	1	523	0.7778	0.0947	0.0982	0.1030
C12	3	523	0.8538	0.4523	0.4843	0.5049
C13	1	522	0.6178	0.0687	0.0759	0.0839
C14	3	508	0.7910	0.4355	0.4757	0.5159

3. Validation of the engine

- 524 contracts: the engine reproduces the published Part C summary within half a star for 98.1% and exactly for 78.1%.
- Not modelled: categorical adjustment index and reward factor (explains the small positive gap).
- 32.4% of MA contracts are rated 3.5 stars (the only ones one boundary crossing can lift to 4.0).

4. Which assumption matters most

Assumption	Provenance	Value At Low Input	Value At High Input	Swing Usd
eligible_share (diabetes / hypertension)	Public source (CMS/KFF)	131.4515	200.8712	69.4197
qbp_per_enrollee_year	Public source (CMS/KFF)	131.1576	192.1995	61.0419
rho_boundary_density (CMS 2026 data)	Public data, computed (CMS 2026 Star Ratings)	146.3134	166.3828	20.0694

5. A plan's own eligible share

The model accepts a plan-supplied share (plan_eligible_share in the configuration, for example diabetes 0.40 for an SNP-heavy plan). A larger share dilutes each closure, so value falls and the break-even lift rises by the scale shown. Mean over the treated gap mix, base QBP and boundary density.

Group	Eligible Share	Mean Usd Per Closure	Break Even Scale
diabetes	0.2000	194.0454	0.8098
diabetes	0.2600	157.1416	1.0000
diabetes	0.3200	134.0768	1.1720
diabetes	0.4000	114.0873	1.3774
diabetes	0.5000	98.0956	1.6019

6. Value by contract star position

- The headline averages over all 524 rated contracts. A plan faces its own position, and the value is concentrated: only a contract rated 3.5 stars can be lifted over the 4-star QBP boundary through a measure.
- Contracts rated 3.0 or less, 4.0 or more, or 3.5 with the measure far from its next cut point get $0 from this route. A 3.5-star contract with the measure near a cut point gets about six times the average.
- Not modelled: a 4.0-star contract near its lower edge protecting its bonus (downside). It would add value for that position. The raw rating inside a half-star band is not published, so it is assumed uniform.

Position	N Contracts	Diab Eye Exam	Diab Nephro Screen	Diab A1c Test	Diab Blood Sugar	Htn Pcp Followup	Mean Over Gap Mix
all_contracts	523	143.9000	111.2000	303.0000	303.0000	238.9000	157.1000
rated_3.0_or_less	120	0	0	0	0	0	0
rated_3.5	170	442.8000	341.4000	932.1000	932.1000	713.9000	480.3000
rated_3.5_measure_near_cut	65	1,327	1,383	1,802	1,802	1,447	1,108
rated_3.5_measure_far_from_cut	138	0	0	0	0	0	0
rated_4.0_or_more	233	0	0	0	0	0	0

7. InComm metadata versus CMS 2026

InComm's StarsWeight differs from CMS's 2026 weight for eye exam and kidney screening, and COPD spirometry and BH follow-up feed no 2026 Part C measure. The model uses CMS's weights.

Caregapcode	Hedismeasureid	Starsweight	Cms Measure ID	Cms 2026 Weight	Note
BH_POST_DISCHARGE_7D	FUH-7	3	n/a	n/a	not a 2026 Part C Star measure
COPD_SPIROMETRY	SPR	1	n/a	n/a	not a 2026 Part C Star measure
DIAB_A1C_TEST	CDC-A1C	3	C12	3.0000	matches
DIAB_BLOOD_SUGAR	CDC-A1C	3	C12	3.0000	matches
DIAB_EYE_EXAM	CDC-EYE	3	C11	1.0000	InComm weight differs from CMS
DIAB_NEPHRO_SCREEN	KED	3	C13	1.0000	InComm weight differs from CMS

HTN_PCP_FOLLOWUP	CBP	3	C14	3.0000	matches